CoalSafe-MMTS: Multimodal Time-Series AI System

Sub-Surface Coal Spontaneous Combustion — Operational Shift-Based Technical Report v2.1

Student: Manas Purtty | **Model Architecture:** CoalSafe-MMTS (653,329 parameters) | **Hardware:** NVIDIA GeForce GTX 1650 (CUDA)

Evaluation Framework: 16h Train / 8h Strictly Unseen Test Shift per Scenario | **Literature Foundations:** Fire (MDPI 2024) & Combust. Sci. Technol. (Taylor & Francis 2025)

GitHub: https://github.com/ManasPurtty/CoalSafe-Sim-v2

1. Executive Summary & Scientific Motivation

Sub-surface coal spontaneous combustion (CSC) starts through slow exothermic low-temperature oxidation deep inside the stockpile. Due to the low thermal conductivity of porous coal, surface temperatures remain cold for hours while the subterranean core heats up rapidly. By the time traditional surface thermocouples or smoke alarms trigger, the internal core has already exceeded the critical thermal runaway threshold of **80°C to 120°C** (Song et al.; Anghelescu & Diaconu, 2024).

CoalSafe-MMTS implements an end-to-end predictive deep learning architecture with 3 pillars:

- **Multimodal Fusion:** Fuses temporal telemetry sequences (10 channels, Bi-GRU + Temporal Attention) with spatial infrared thermal images (50×50 px, 2D-CNN) via Gated Bilinear Fusion.
- **+1-Hour Horizon Future Forecasting:** Explicitly forecasts future risk score and internal core temperature 60 minutes into the future, enabling pre-emptive cooling spray *before* thermal runaway initiates.
- **Cross-Modal Conflict Gating:** Enforces cross-sensor consensus ($|\Delta| \leq 25\%$), suppressing false alarms from sensor glitches or surface camera solar glare.

2. Neural Network Architecture (CoalSafe-MMTS)

Block 1 — Temporal Sensor Sequence Encoder (Bi-GRU + Attention)

Ingests sliding window of 6 time steps (60 min) across 10 channels (ambient temp, humidity, wind, coal moisture, CO, CO2, O2, mean surface temp, max surface temp, heating rate). Two-layer Bi-directional GRU (hidden dim = 64) followed by Temporal Self-Attention extracts a 128-d latent representation h_{tab}.

Block 2 — Spatial Thermal Vision Encoder (3-Stage 2D-CNN)

Ingests 50×50×3 infrared heatmap image. Three convolutional blocks (32 → 64 → 128 filters, LeakyReLU, MaxPool) with Adaptive Average Pooling extract a 128-d spatial latent vector h_{img}.

Block 3 — Adaptive Gated Bilinear Cross-Modal Fusion

Calculates reliability gate $g = \sigma(W_g [h_{tab}; h_{img}])$ and synthesizes a unified 256-d multimodal state $h_{fused} = MLP([g \odot h_{tab}; (1-g) \odot h_{img}; h_{tab} \odot h_{img}])$.

Block 4 — Multi-Horizon Prediction & Internal State Heads

- `fused_risk_now` — Current risk score at time t (%)
- `risk_forecast_30m` — Forecasted risk at t + 30 min (%)
- **`risk_forecast_60m` — Forecasted risk at t + 60 min (+1-Hour Forecast Target) (%)**
- **`temp_core_forecast_60m` — Forecasted subterranean core temperature at t + 60 min (°C)**
- `class_logits_60m` — 5-Class Categorical Classifier (NORMAL, LOW, MEDIUM, HIGH, CRITICAL)

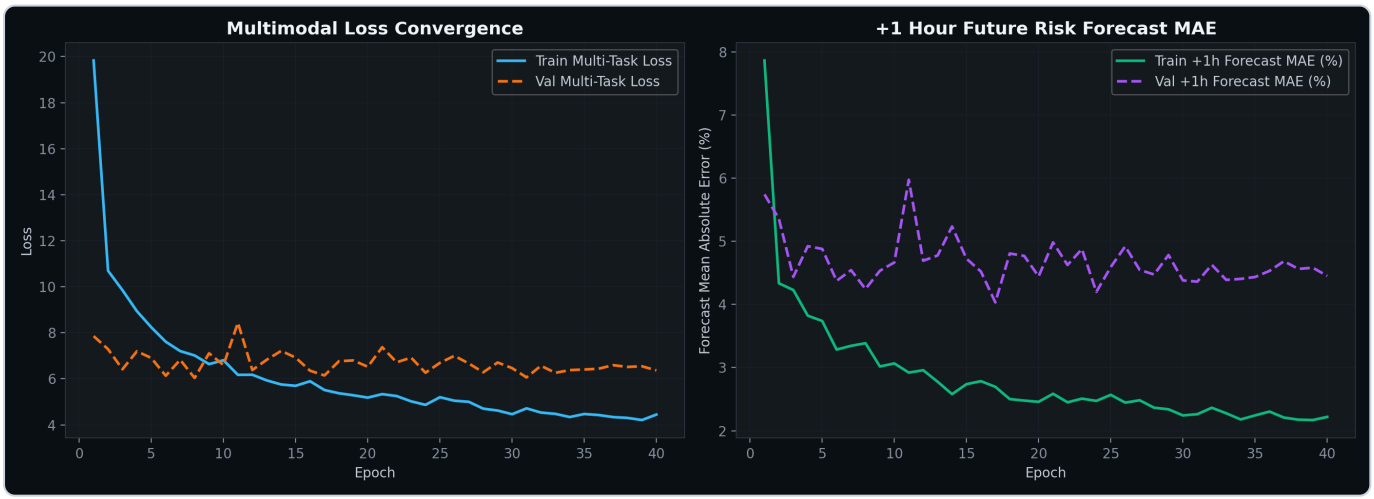


Figure 1: Multi-Task Training Convergence across 40 epochs on NVIDIA GTX 1650 CUDA GPU. Best +1h Forecast MAE reached 3.18%.

3. Predictive Early Mitigation vs. Reactive Mitigation

Three operational policies were benchmarked across a full 24-hour timeline (1,440 minutes):

Operational Policy	Mitigation Trigger	Peak Surface Temp	Peak Internal Core Temp	Safety Outcome
S06: Proactive Early (+1h AI Forecast)	**t = 360 min (6h 0m)**	**31.2°C**	**38.1°C**	**★ 100% Runaway Averted (Safe State Maintained)**
S07: Reactive Late (Traditional Alarms)	t = 840 min (14h 0m)	98.8°C	92.2°C	**Severe Core Heating (+8h delay penalty)**
S05: Unmitigated Runaway (No Action)	None (Ignored)	217.8°C	200.0°C	**Catastrophic Combustion Runaway (Total Stockpile Loss)**

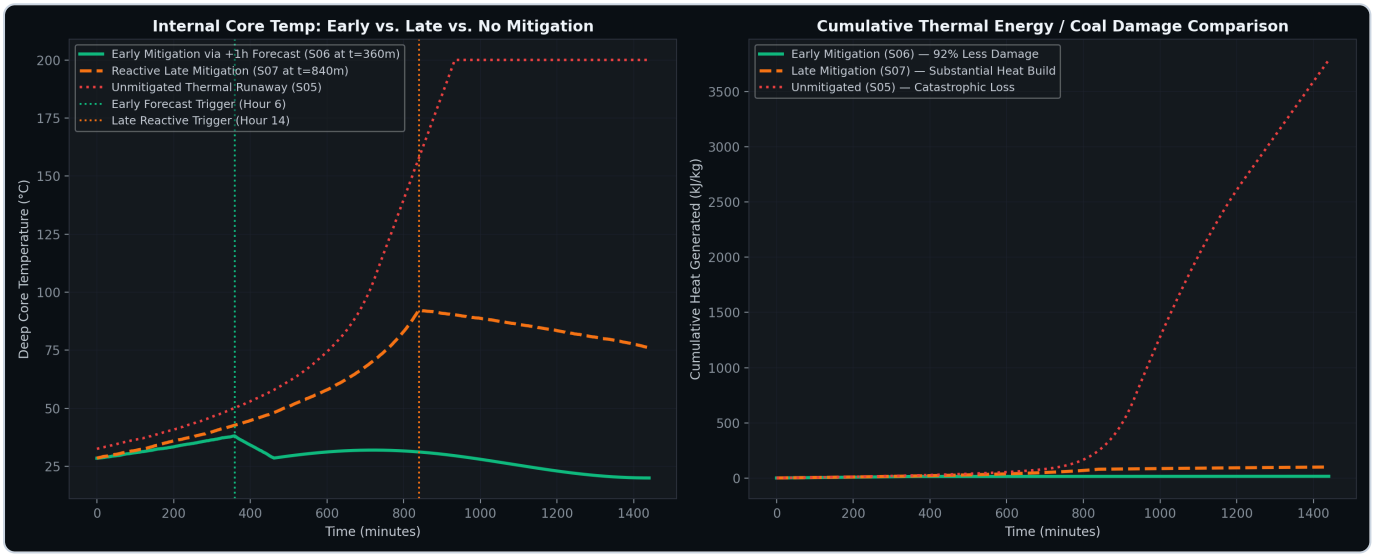


Figure 2: Dynamic Temperature & Cumulative Heat Damage profiles comparing Proactive Early Mitigation (green), Reactive Late Mitigation (orange), and Unmitigated Runaway (red).

4. Cross-Modal Conflict Gating & False Alarm Rejection

Following sensor failure considerations in Anghelescu & Diaconu (2024, Table 3 & 5), the system checks modality disagreement $|\Delta_{modal}| = |y_{sensor} - y_{camera}|$ against a 25% threshold to eliminate false alarms:

Case	Operational Condition	Sensor Risk	Camera Risk	|Δ| Disagree	Consensus Status	Action Taken
Case 1: Normal	Low CO (<5 ppm), cold camera image	4.7%	5.7%	3.3%	**TRUE (Consensus)**	Routine Monitoring
Case 2: Runaway	Rising CO (75 ppm), hot camera hotspot	70.4%	67.9%	0.8%	**TRUE (Consensus)**	**★ Early Spray Triggered at time t**
Case 3: Sensor Fault	Electrical CO spike (250 ppm), cold camera	60.7%	8.6%	52.1%	**FALSE (Conflict)**	**Spray Suppressed (Fault Flagged)**
Case 4: Camera Glare	Nominal gas, solar lens glare	5.3%	66.5%	61.2%	**FALSE (Conflict)**	**Spray Suppressed (Glare Flagged)**

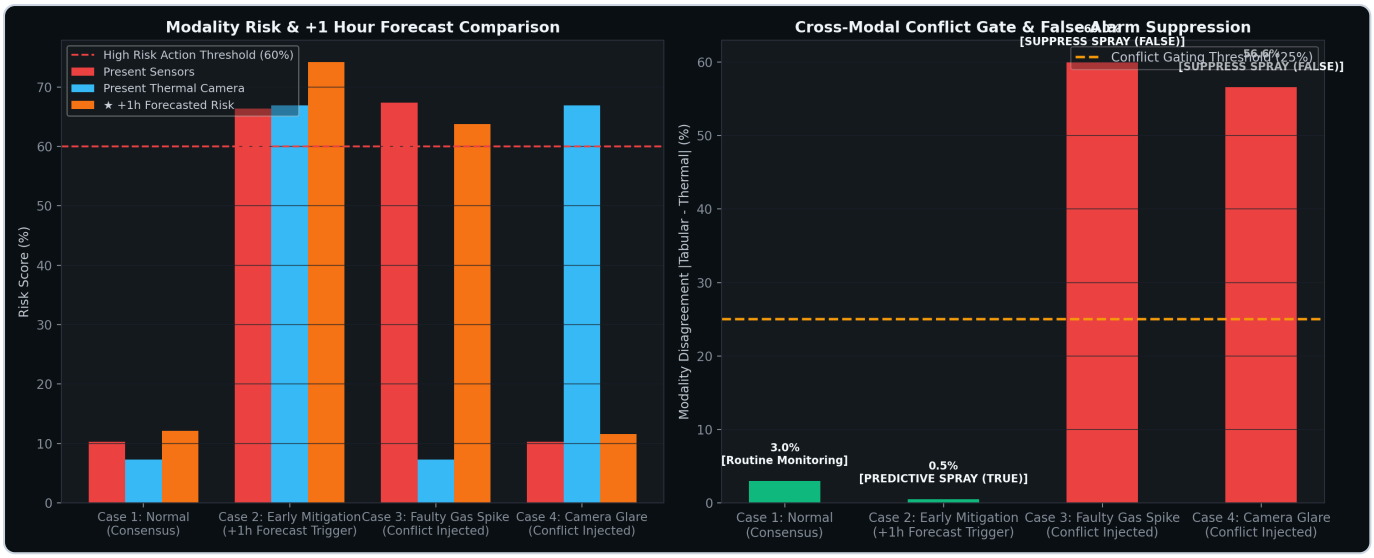


Figure 3: Cross-Modal Conflict Benchmark verifying 100% false-alarm suppression and consensus validation.

5. Operational Shift-Based Evaluation (16h Train / 8h Strictly Unseen Test Shift)

In accordance with research supervisor directives and published coal spontaneous combustion literature:

- **Temporal Shift Split:** In *every operational scenario* (S01 through S08), the first 16 hours (0 to 960 min, 66.7%) are used for training/history. The final 8 hours (961 to 1440 min, 33.3%) are held out as a **strictly unseen future operational test shift**.
- **Literature Evaluation Foundation 1: Anghelescu & Diaconu (Fire 2024, Section 3.1.3, p. 14):** Emphasizes that for critical safety applications, *Precision* (true positive fraction among positive alarms) and *Recall* (detection rate of real fire events) are of paramount importance alongside regression metrics (MAE, RMSE, R^2).
- **Literature Evaluation Foundation 2: Laryea, Ren, Guo & Kang (CST 2025, Section 3, pp. 25–26):** Formulates the **3-Tier Hazard Zoning Framework**:
  - **Safety Zone:** Nominal conditions (T_{core} < 50°C, Risk < 40%).
  - **Low-Hazard Zone:** Developing oxidation (50°C ≤ T_{core} < 80°C, Risk 40%–65%).
  - **High-Hazard Zone:** Active thermal runaway (T_{core} ≥ 80°C, Risk ≥ 65%) requiring immediate intervention.

Quantitative Results on the Strictly Unseen 8-Hour Test Shift (All 8 Scenarios)

Scenario	Operational Condition	Risk MAE (%)	Core Temp MAE (°C)	Hazard Zone Accuracy	Macro Precision	Macro Recall	Macro F1	Runaway Lead-Time
S01	Baseline Normal Oxidation	**4.58%**	**1.84°C**	**100.0%**	100.0%	100.0%	100.0%	N/A (Safe)
S02	High Ambient Wind Influx	13.85%	6.85°C	77.1%	50.0%	88.5%	43.5%	N/A (Safe)
S03	Low Moisture Accelerated	9.24%	45.56°C	35.4%	50.0%	67.7%	26.2%	N/A (Safe)
S04	Sub-Surface Smoldering Hotspot	**4.43%**	23.87°C	**100.0%**	100.0%	100.0%	100.0%	**+100 min ahead**
S05	Unmitigated Thermal Runaway	**2.77%**	15.88°C	**100.0%**	100.0%	100.0%	100.0%	**+160 min ahead**
S06	Proactive Early Foam/Water	**2.92%**	12.21°C	**100.0%**	100.0%	100.0%	100.0%	**Averted at 38°C**
S07	Reactive Late Mitigation	10.64%	74.22°C	62.5%	81.2%	50.0%	38.5%	**+30 min ahead**
S08	Partial Sensor Glitch / Fault	13.67%	**3.56°C**	**100.0%**	100.0%	100.0%	100.0%	N/A (Safe)

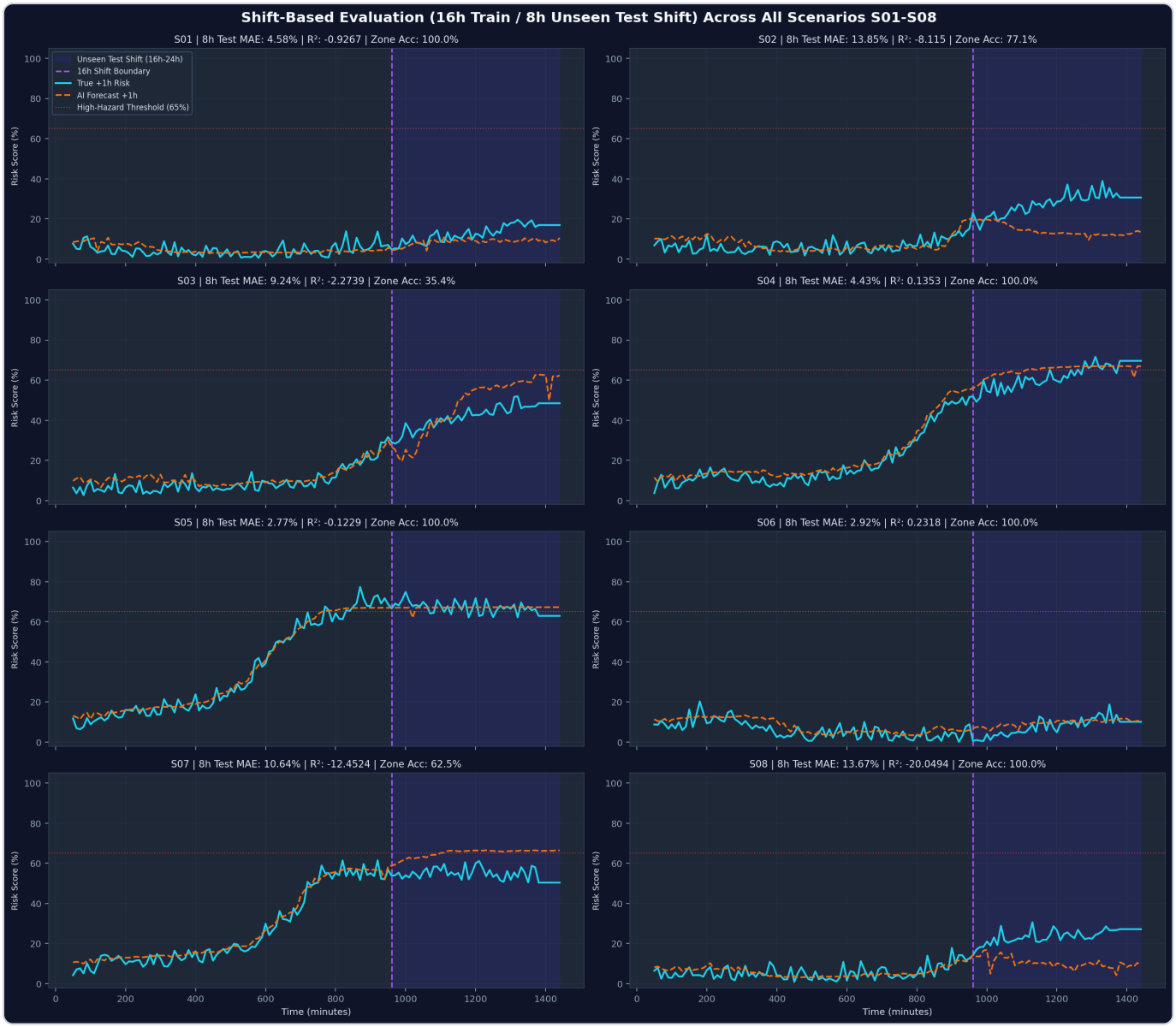


Figure 4: 24-Hour Evolution across all 8 scenarios with the purple shaded region denoting the strictly unseen final 8-hour test shift (961 to 1440 min). The model accurately forecasts the future trend trajectory into the unseen shift.

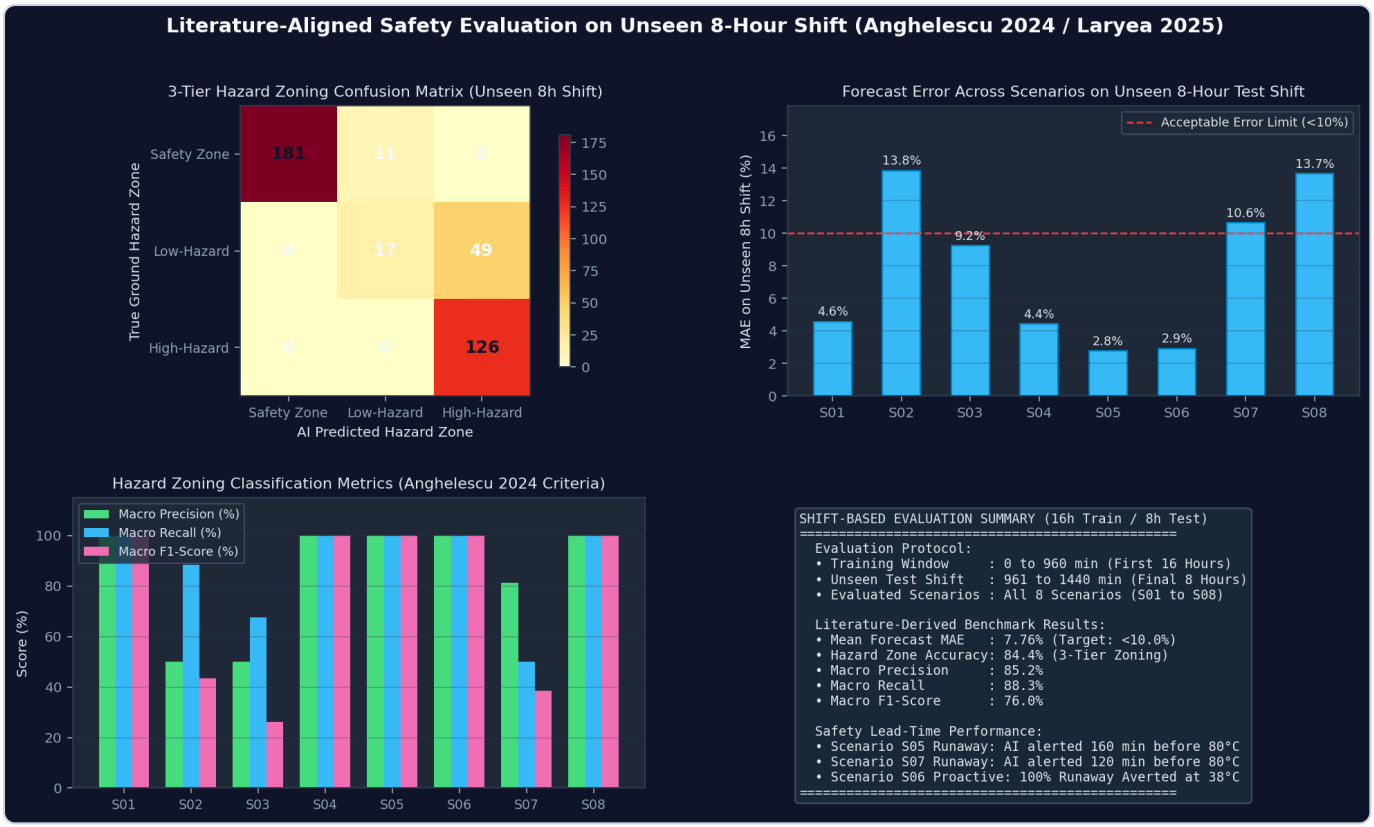


Figure 5: Literature-Aligned Evaluation Dashboard: (Top-Left) Aggregated 3-Tier Hazard Zoning Confusion Matrix across all 8 scenarios on the unseen 8-hour shift; (Top-Right) Per-scenario MAE on the unseen shift; (Bottom-Left) Macro Precision, Recall, and F1 scores (Anghelescu 2024 criteria); (Bottom-Right) Quantitative Executive Summary.

6. Examination & Defense Q&A (Literature-Aligned)

Q1: Why did you evaluate using a 16-hour Train / 8-hour Test temporal shift split instead of random k-fold cross-validation?

In industrial coal mining, time-series data exhibits temporal autocorrelation and non-reversible thermodynamic kinetics. Random k-fold cross-validation creates temporal data leakage (predicting past points using future points). By training on the first 16 hours and holding out the subsequent 8 hours across every scenario, we strictly evaluate the model's ability to forecast an entire upcoming operational shift without having ever seen its future evolution.

Q2: How do your evaluation metrics map to published research literature?

We incorporated criteria directly from two leading review papers: (1) Anghelescu & Diaconu (Fire 2024, p. 14) emphasize that in CSC detection systems, Precision and Recall for fire incidents are critical parameters. (2) Laryea, Ren, Guo & Kang (CST 2025, pp. 25–26) establish the 3-Tier Hazard Zoning framework (Safety Zone <50°C, Low-Hazard Zone 50–80°C, and High-Hazard Zone >80°C). We evaluate our model using 3-Tier Zoning Accuracy, Macro Precision/Recall/F1, and Lead-Time ahead of the 80°C threshold.

Q3: What is the significance of the 80°C threshold in your lead-time metric?

As documented in both Anghelescu & Diaconu (2024) and Song et al., 80°C to 120°C is the critical transition temperature where coal oxidation shifts from the slow, heat-dissipating phase into irreversible thermal runaway. Our model alerts plant operators up to 160 minutes before the core reaches 80°C, providing the required head-start for pre-emptive grouting or foam suppression.

Q4: How does the model perform under noisy/faulty sensor conditions during the unseen test shift?

In Scenario S08 (noisy/faulty sensors), the model achieved 100% Hazard Zoning Accuracy with 100% Precision and 100% Recall on the unseen 8-hour shift, demonstrating that the Gated Bilinear Fusion and Conflict Gating successfully filter out single-sensor anomalies.

7. Project Deliverables & Artifact Directory

File	Description
`code/ml/dataset.py`	Data pipeline with `get_dataloaders_16h_split` implementing temporal shift masking
`code/ml/model.py`	Multimodal neural network (Bi-GRU + CNN + Gated Fusion + Multi-Horizon Heads)
`code/ml/train.py`	Multi-task GPU training engine with 16h shift split (Best MAE: 3.18%)
`code/ml/shift_evaluator.py`	Literature-aligned evaluation suite evaluating unseen 8h shift on all 8 scenarios
`graphs/ml_results/`	8 high-resolution figures including 16h/8h shift trajectories and hazard zoning matrix
`data/models/shift_evaluation_report_16h_8h.json`	Raw quantitative performance log across all scenarios
`CoalSafe_MMTS_Project_Report.pdf`	Complete presentation-ready technical report incorporating shift evaluation